

15	For a vertical oscillating spring-mass system, which of the following changes will not decrease the resonance frequency of the system?	
	A	Decreasing the amplitude of the oscillation
	B	Decreasing the stiffness of the spring used
	C	Increasing the mass of the oscillating object
	D	Increasing the resistive forces acting on the system
	Answer: A	

The amplitude of the oscillation has no effect on the natural frequency and hence does not affect the resonance frequency of the system.

Decreasing k and increasing m will decrease the natural frequency of the oscillation since

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}.$$

Increasing the damping will also decrease the resonance frequency of the system.